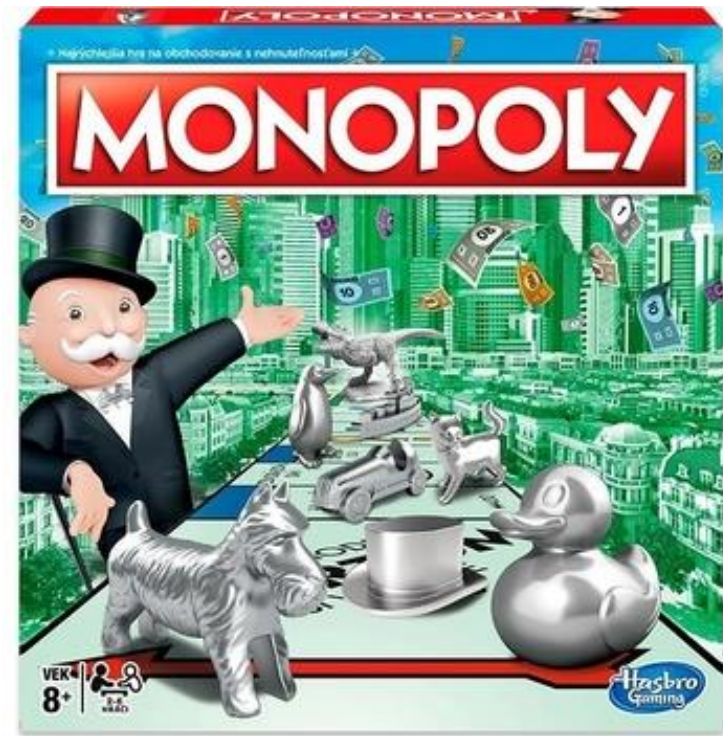


Microeconomics II



Topic 7: Monopoly

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Examples

Google



Roadmap

1. Profit maximisation under monopoly
2. Inefficiency of monopoly
3. What causes monopolies?

Aims of the lecture

Students (you) can explain

1. How a monopoly maximises profits and how its marginal revenues differ from firms under perfect competition.
2. How monopoly pricing is related to the elasticity of demand.
3. How market power is measured.
4. Several different factors that give rise to monopolies.
5. Why regulating monopolies is challenging.

Motivation



Roadmap

1. Profit maximisation under monopoly
2. Inefficiency of monopoly
3. What causes monopolies?

Profit maximisation of monopoly

- Problem: $\max \pi(y) = r(y) - c(y) = p(y)y - c(y)$
- FOC: $MR(y^*) = MC(y^*)$
 - Perfect competition: $MR(y^*) = p$
 - Monopoly: $MR(y^*) < p$
- SOC: $\frac{d^2\pi(y^*)}{dy^2} = \frac{dMR(y^*)}{dy} - \frac{dMC(y^*)}{dy} < 0$
 - In optimum, MC curve has a steeper slope than MR curve

Profit maximisation: Exercise

Setup:

- Market demand: $D(p) = 20 - \frac{p}{4}$
- Total costs: $c(y) = 16y + 200$

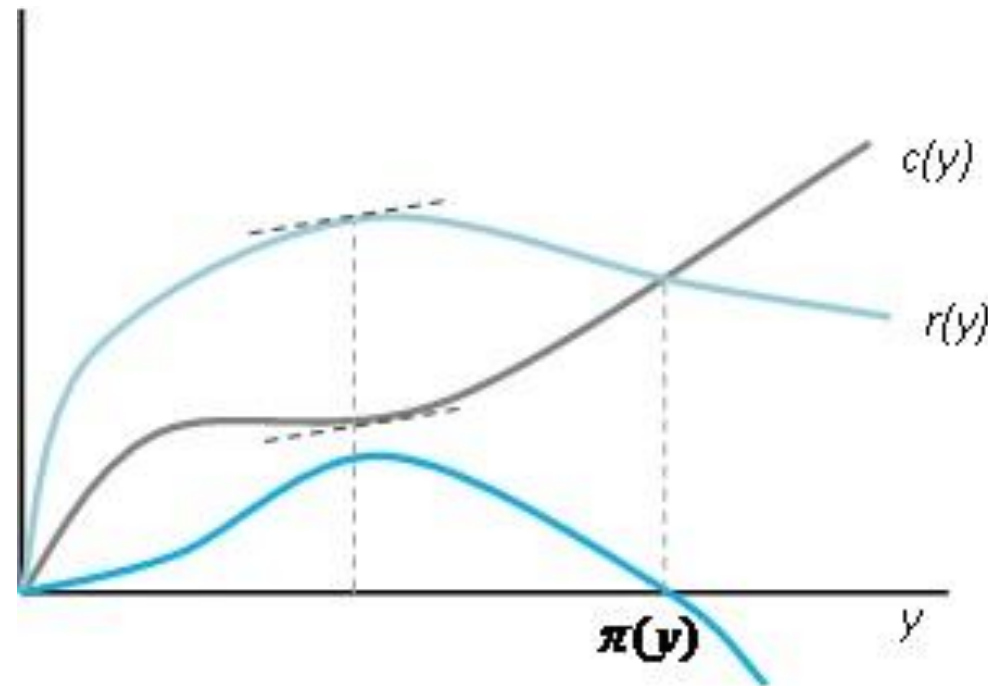
Question:

- What is the optimum output?
- What price will the monopoly charge?
- What will its profit be?



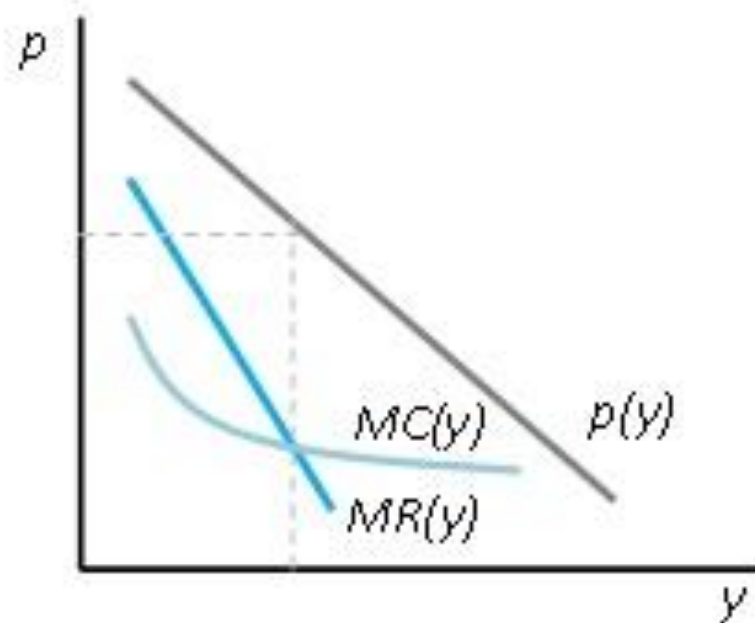
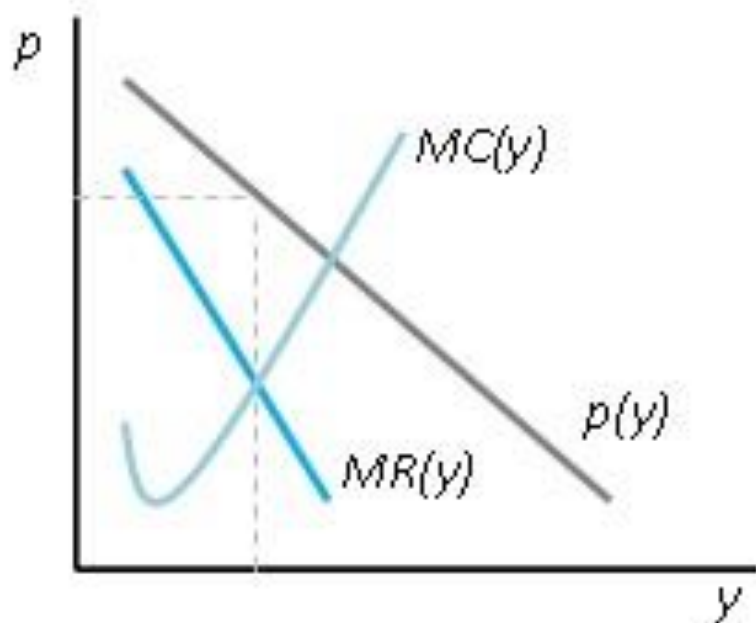
Profit maximisation: First order condition

$$MR(y^*) = MC(y^*)$$



Profit maximisation: Second order condition

$$\frac{dMR(y^*)}{dy} < \frac{dMC(y^*)}{dy}$$



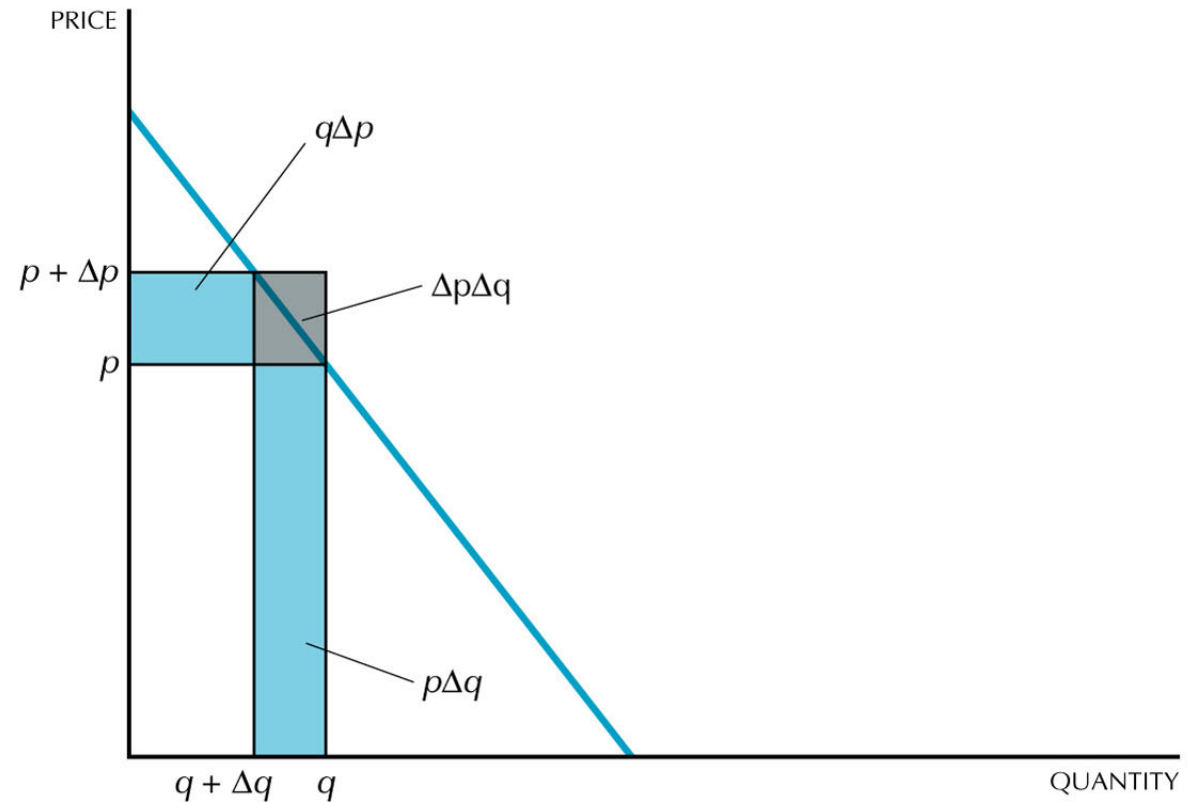
Decomposing the change in revenues

Higher price leads to lower quantity

➔ Revenue may increase or decrease

➔ Importance of elasticity of the demand.

$$\begin{aligned}\Delta R &= R' - R \\ &= (p + \Delta p)(q + \Delta q) - pq \\ &= p\Delta q + \Delta p q + \Delta p \Delta q \\ &\approx p\Delta q + \Delta p q\end{aligned}$$



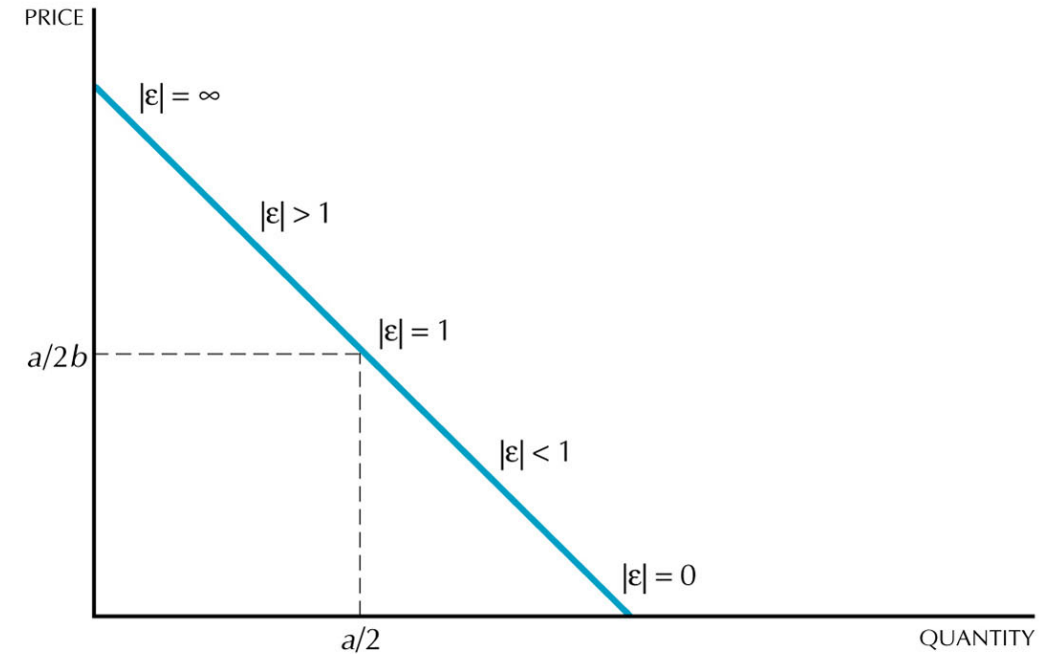
Change in revenue with price change

$$\frac{\Delta R}{\Delta p} = q + p \frac{\Delta q}{\Delta p}$$

Positive if $\frac{\Delta q}{\Delta p} p > -1$, i.e. $|e(p)| < 1$

➔ If $|e(p)| < 1$, monopoly could increase revenue and at the same time reduce costs by increasing price (and reducing output)

➔ Monopoly always selling on the elastic part of demand curve



Change in revenue with output change = MR

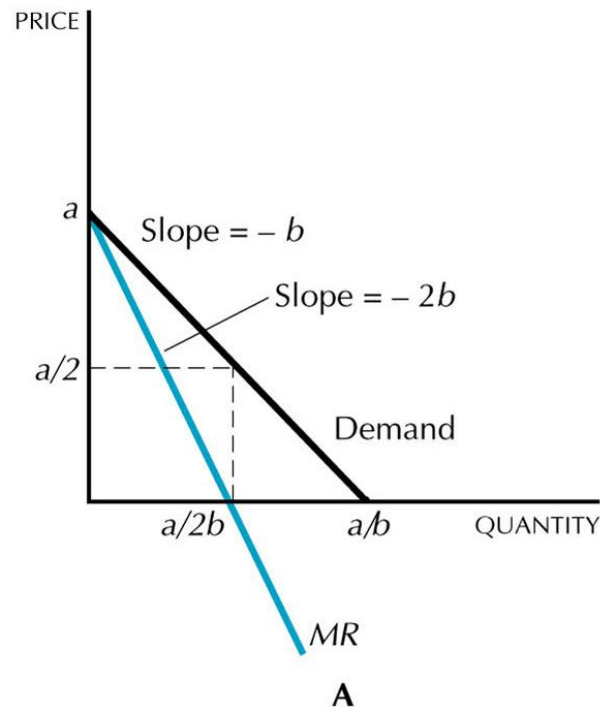
$$\begin{aligned}\text{MR} &= \frac{\Delta R}{\Delta q} \\ &= p + q \frac{\Delta p}{\Delta q} \\ &= p \left(1 + \frac{q}{p} \frac{\Delta p}{\Delta q} \right) \\ &= p(q) \left(1 - \frac{1}{|e(q)|} \right)\end{aligned}$$

- ➔ Increasing output increases revenues if $|e(q)| > 1$
- ➔ Again, monopoly will always sell on the elastic part of the demand curve

MR with two common types of demand curves

Linear demand curve

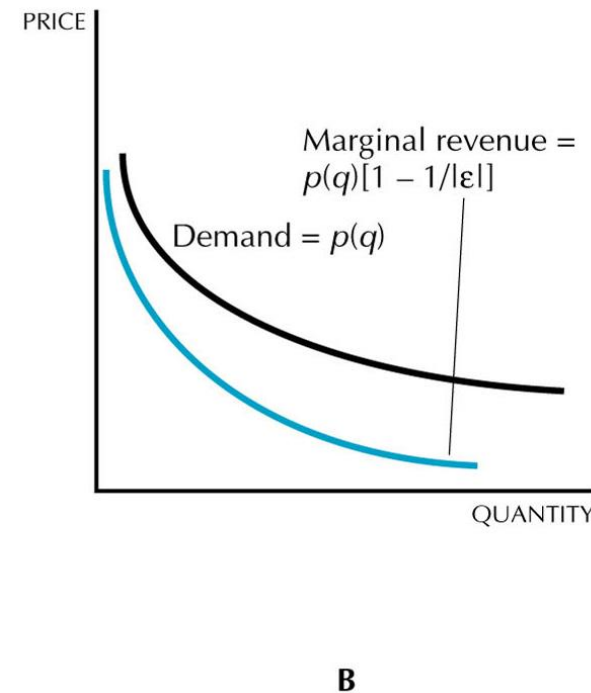
- $p(q) = a - bq$
- $TR(q) = p(q)q = aq - bq^2$
- $MR(q) = a - 2bq$



Constant elasticity demand curve

- $MR(q) = p(q)(1 - \frac{1}{|e|})$

(NOTE: given by $q = Ap^e \rightarrow p(q) = \ln_{-e} A - \ln_{-e} q$)



Markup

$$MC(y) = MR(y) = p(y) \left(1 - \frac{1}{|e(y)|} \right)$$

$$\rightarrow p(y) = \frac{MC(y)}{1 - \frac{1}{|e(y)|}}$$

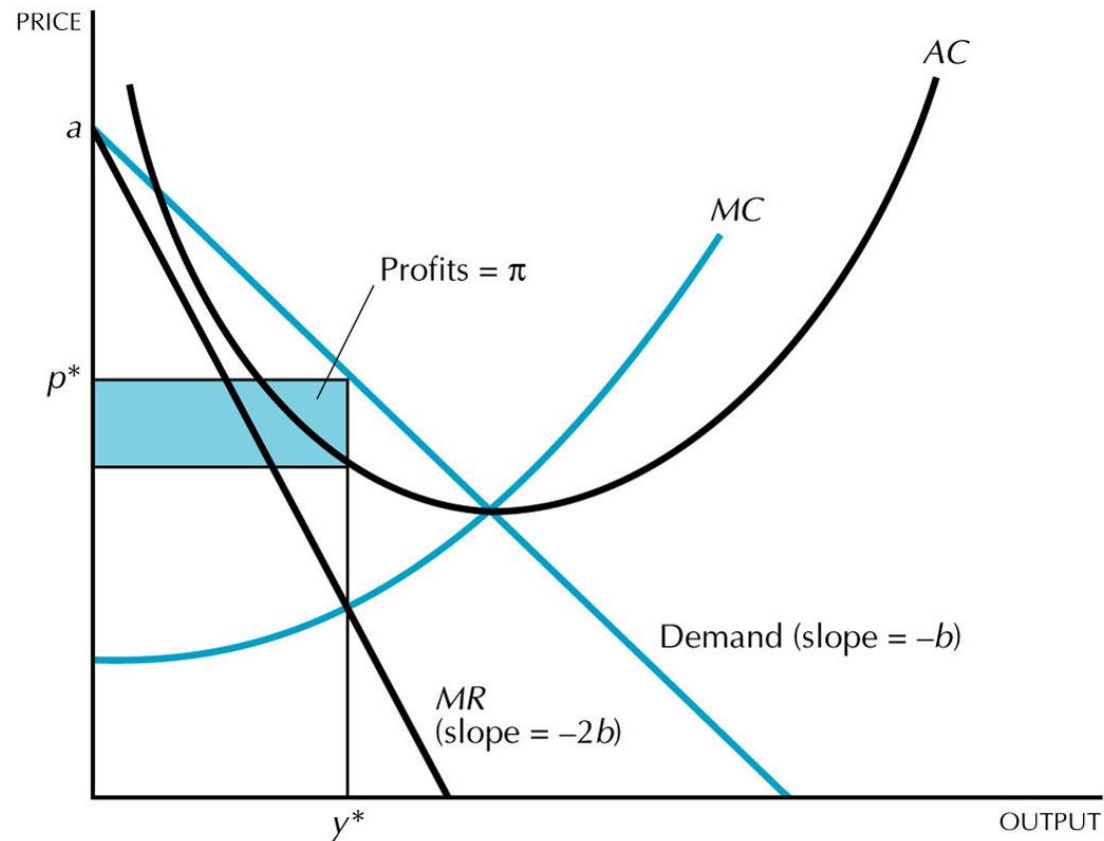
→ Monopolist charges price greater than marginal costs

Markup

- We can write $p(y) = \mu MC(y)$, where $\mu = \frac{1}{1 + \frac{1}{e(y)}}$ is markup
- $\mu = \frac{p}{MC}$
- $|e(y)| > 1 \rightarrow \frac{1}{1 + \frac{1}{e}} > 1 \rightarrow$ monopoly markup greater than 1
- Decreasing with $|e(y)|$
- Often used as a measure of market power

Profit of a monopolist

- Optimal output y^* : given by $MR(y) = MC(y)$
- Optimal price p^* : the highest possible at optimal output y^*
- Profit: $\pi(y) = y(p(y) - AC(y))$



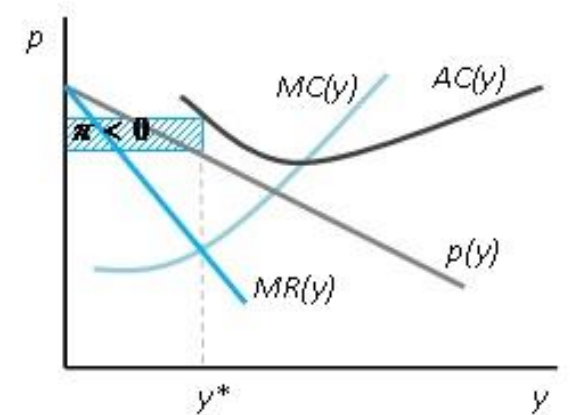
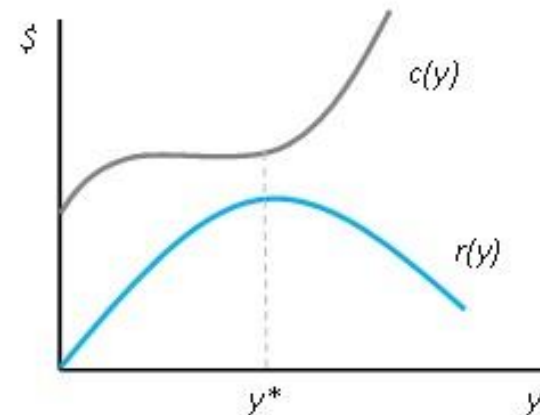
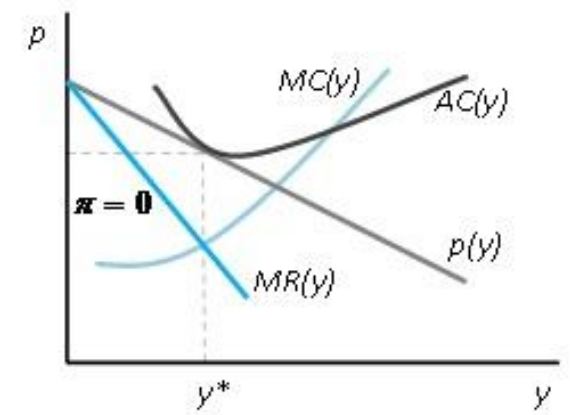
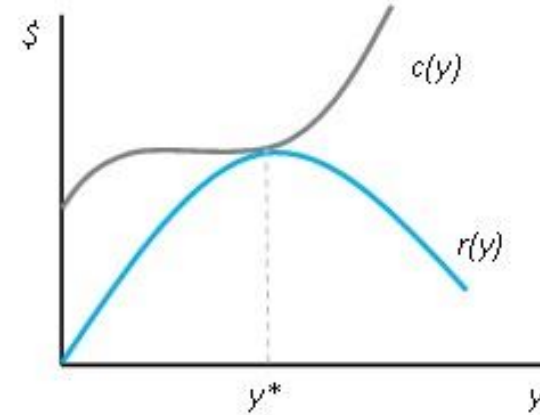
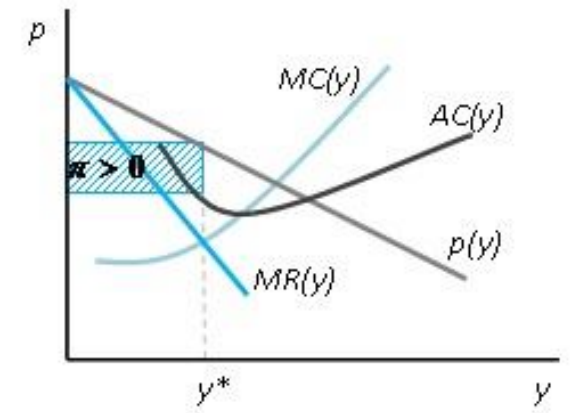
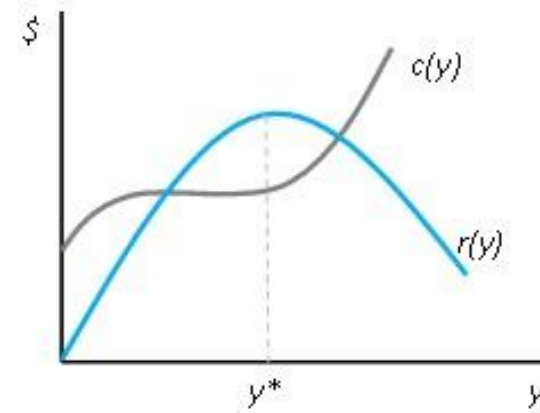
Profit of a monopolist

Typically assumed that monopoly has high profits. But...

- Price above marginal cost, but below average cost → negative profit
- Short-run: firm willing to produce with a loss, if it is not greater than the fixed cost, i.e. price is not below average variable cost
- Long-run: exit, if no perspective of positive development (may stay if expected profits high and costly to enter back)



Profit of a monopolist

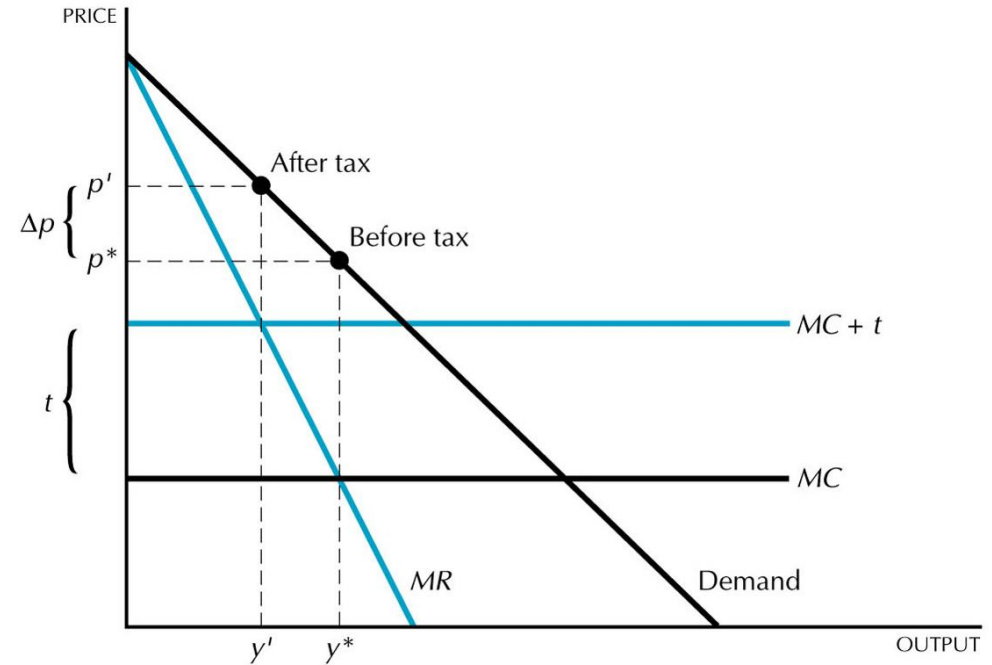


The effect of taxes

Are taxes passed to consumers?

Constant MC + linear demand curve

- Tax moves firm to more elastic part of the demand curve
- Only part of the tax passed to consumers



Constant elasticity demand

- $p = \frac{MC}{1 - \frac{1}{|e|}} = \frac{c+t}{1 - \frac{1}{|e|}}$
- $\frac{\Delta p}{\Delta t} = \frac{1}{1 - \frac{1}{|e|}} > 1$

➔ Monopolist passes on more than the amount of the tax

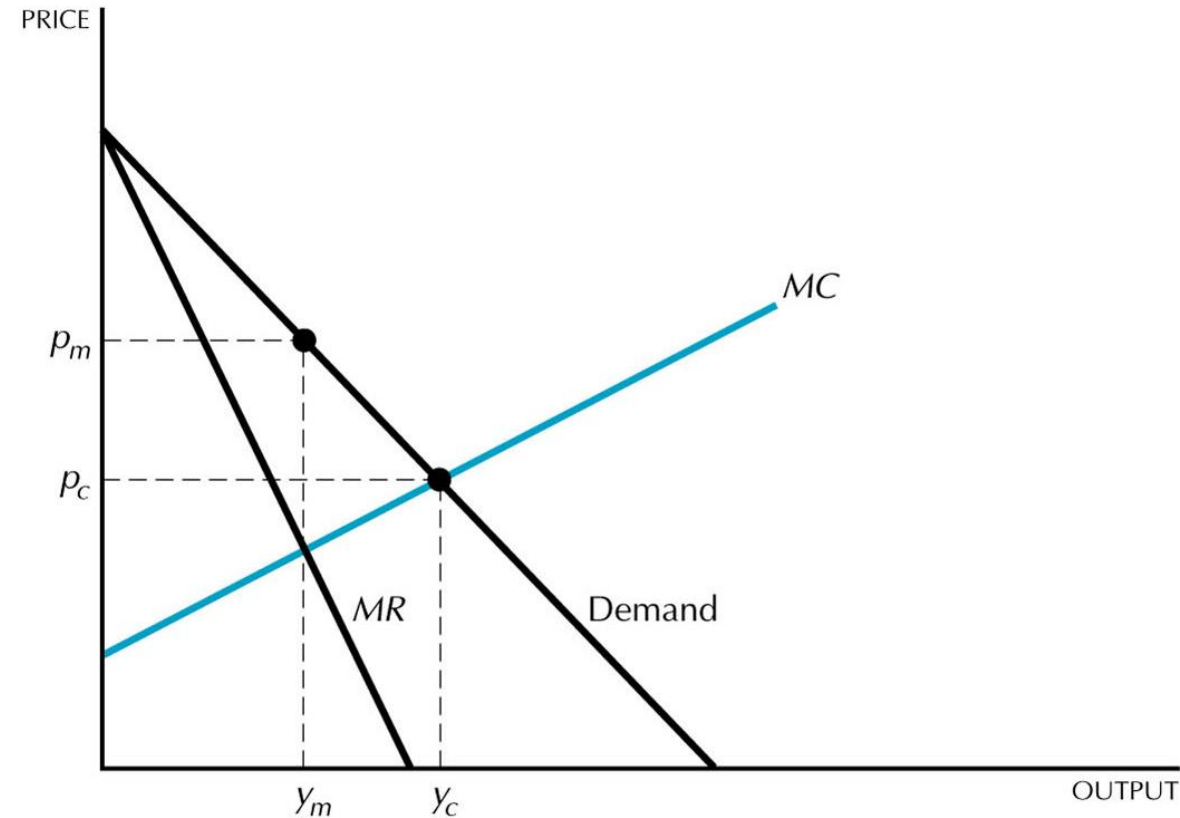
Roadmap

1. Profit maximisation under monopoly
2. Inefficiency of monopoly
3. What causes monopolies?

Inefficiency of monopoly

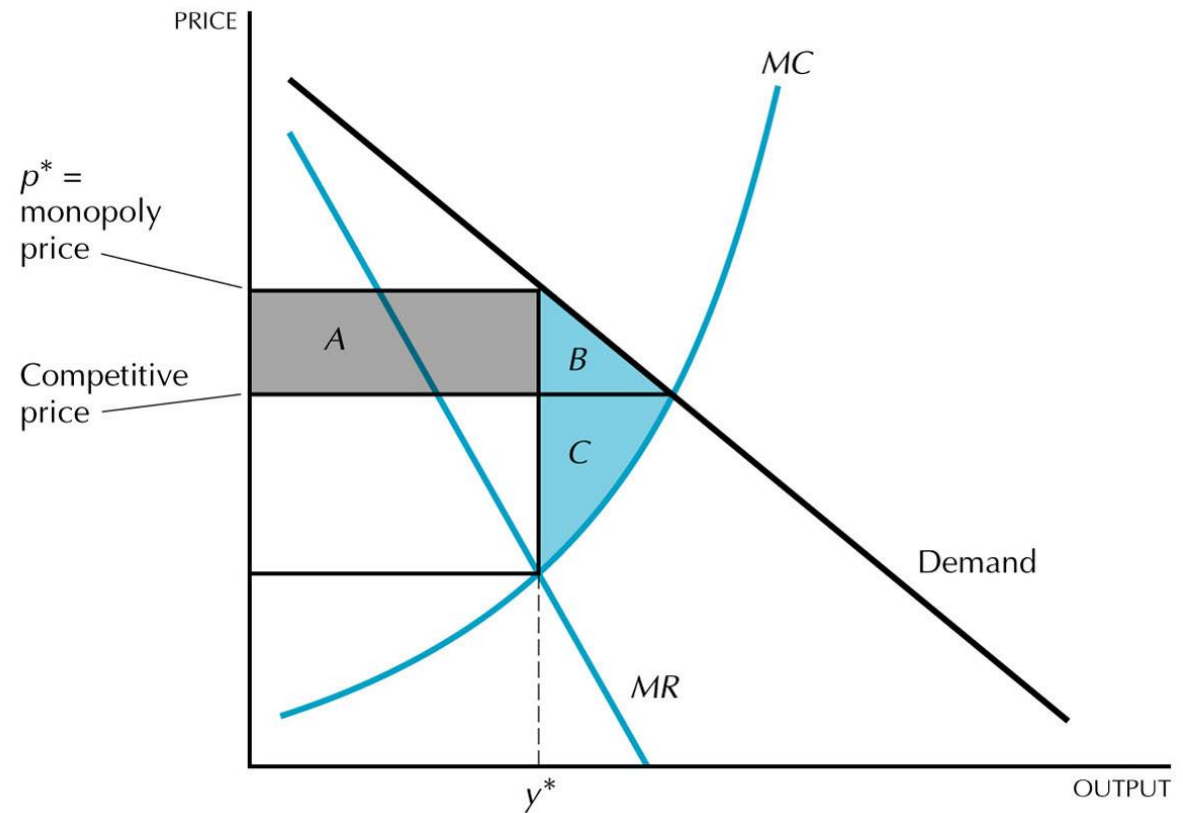
Compared to perfect competition

- Higher price
- Lower output
- Consumers worse off
- Firm better off
- Pareto inefficient: people willing to pay more for additional units that it costs to produce them



Inefficiency of monopoly: Dead-weight loss

- Treat consumers and producer symmetrically.
- Change in producer's surplus A-C
- Change in consumers' surplus $-(A+B)$
- Deadweight loss $-(B+C)$



Inefficiency of monopoly: Exercise

Suppose you are in charge of a toll bridge that is essentially cost-free. The demand for bridge crossings, Q , is given by $P = 12 - 2Q$.

1. Draw the demand curve for bridge crossings
2. What is the socially optimal price for crossing the bridge?
How many people will cross the bridge at that price?
3. If you were a monopolist, what price would you charge?



Measuring market power of monopoly

Market concentration

1. Concentration ratio

- Definition: share of k largest firms to the total industry output
- Problem: does not capture concentration within top k or outside top k

2. Herfindahl index:

- Definition: $H = \sum s_i^2$, where s_i is the market share of firm i
- Zero for pure competition, 10,000 for monopoly.
- Problem: Different distributions of market shares can give the same H
- Advantage: Relatively straightforward to measure
- Problem: Even a market with a large market share can behave competitively (e.g. due to threat of entry)

Measuring market power directly

- Two interchangeable measures:

1. **Markup**: $\mu = \frac{p}{MC} = \frac{1}{1+\frac{1}{e}}$ (1 for perfect competition, >1 for monopoly, goes to ∞ as e goes to -1)

2. **Lerner index**: $LI = \frac{p-MC}{p} = -\frac{1}{e}$ (0 for perfect competition, >0 for monopoly, goes to 1 as e goes to -1)

- Problem: More difficult to estimate

Measuring market power of monopoly: Example



Coca-Cola

- Sales thousands times larger
- Coca-Cola drinkers price-sensitive, many substitutes → elasticity = -4.1.
- $LI = -\frac{1}{e} = 0.244$



Dr. Brown's

- Specialty sodas, celery-flavored soda Cel-Ray
- Cel-Ray drinkers have a unique preference for celery flavour → almost no substitutes
→ elasticity = -2
- $LI = -\frac{1}{e} = 0.5$
- Greater market power and higher markup over its marginal cost

→ Market power is not the same as the size of the firm

Roadmap

1. Profit maximisation under monopoly
2. Inefficiency of monopoly
3. What causes monopolies?

What causes monopolies?

1. Control of important inputs (no close substitutes, entry not possible)

- No close substitutes, entry of others not possible.

Example

- In 1800s, all rubber from trees in Brazil
 - Trees could not be planted close together because of the leaf-blight fungus which would easily spread.
- In 1876, 70 thousand rubber seed stolen by the British
 - Concentrated plantations in Malaysia, no fungus there
 - Dramatic reduction of the cost of harvesting rubber.
 - Cost advantage to the British - in early 1900s the plantations were meeting 95% of world's demand for rubber.
- In 1927, Henry Ford needed rubber for car tires
 - Rubber plantation in the Amazon, called Fordlandia
 - Because of the fungus, the plantation failed
 - Britain's market power survived until the development of synthetic rubber.

What causes monopolies?



1. Control of important inputs (no close substitutes, entry not possible)

- E.g. Rare metals (many in China)

Case study

- In 1800s, all rubber from trees, most rubber trees in Brazil
 - Trees could not be planted close together because of the leaf-blight fungus which would easily spread.
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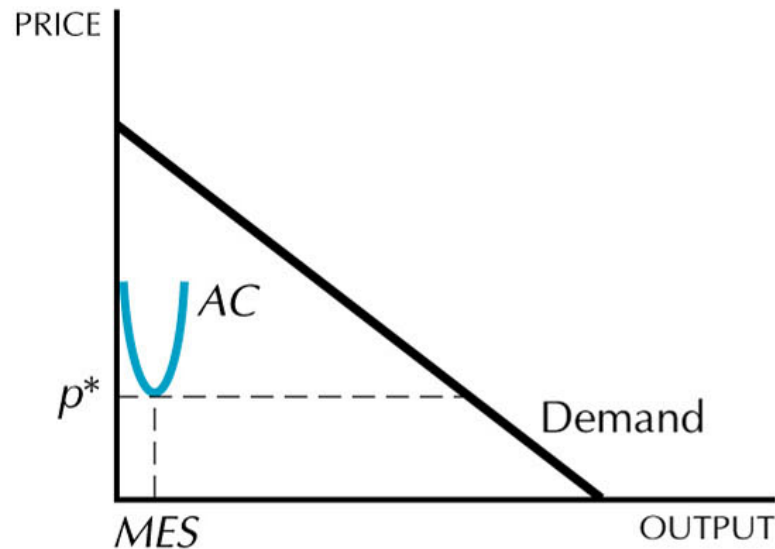
What causes monopolies?



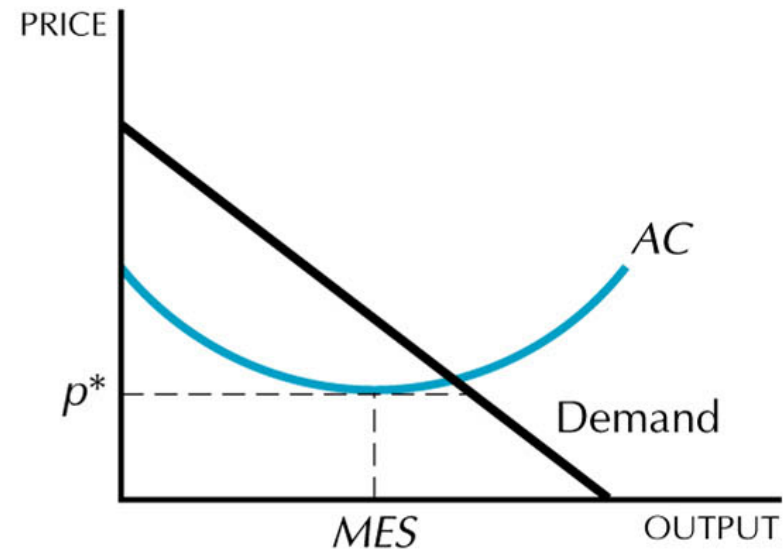
What causes monopolies?

1. Economies of scale (natural monopoly)

- Decreasing AC at output levels important relative to the market demand
- Production by one firm is the least costly



A



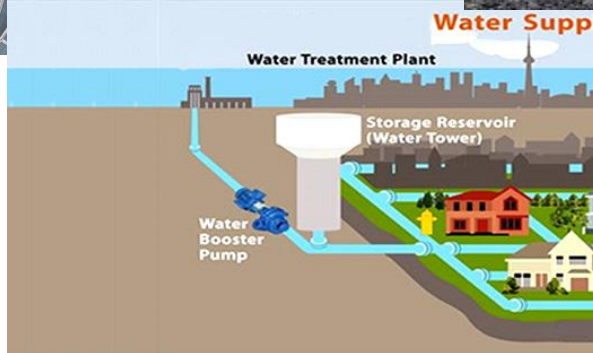
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What causes monopolies?

1. Economies of scale (natural monopoly): Examples



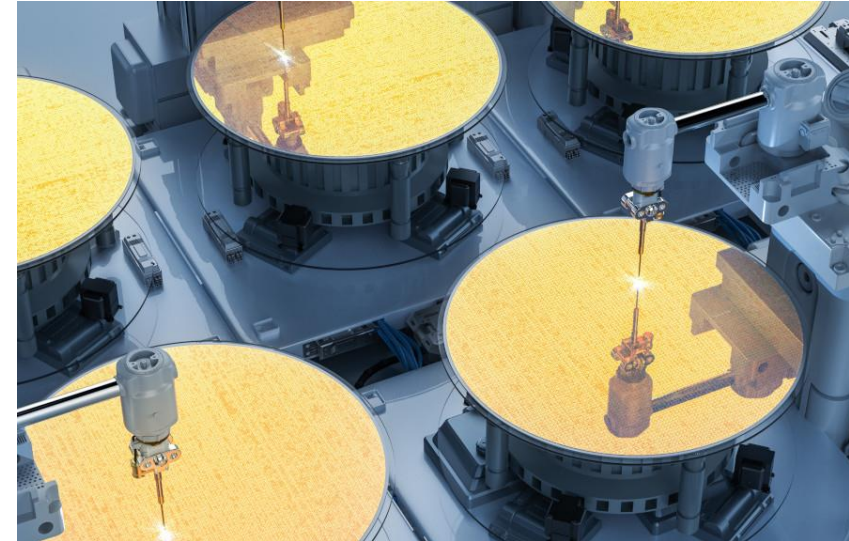
Water Supply & Distribution System



What causes monopolies?

1. Economies of scale (natural monopoly)

- AC curve determined by technology
- But size of the market can be affected by foreign trade policies
- Example: e.g. semi-conductors (a single fab costs USD 20 billion)



What causes monopolies?

2. Control of important inputs (no close substitutes, entry not possible)

- E.g. Mineral water stream; rare metals

Case study

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What causes monopolies?

3. Patents

- Awards exclusive right to benefit from invention
- Aimed to encourage innovation
- What is the most valuable company in Europe?



What causes monopolies?

4. Government licences

- Only firms with a licence are legally allowed to operate in a given market

- Rationales

- Insufficiently defined property rights to a limited input
 - E.g.: spectrum auctions



- Industries characterized by asymmetric information
 - E.g.: education, health, weapons, energy

- Attempt to limit consumption
 - E.g.: government liquor stores



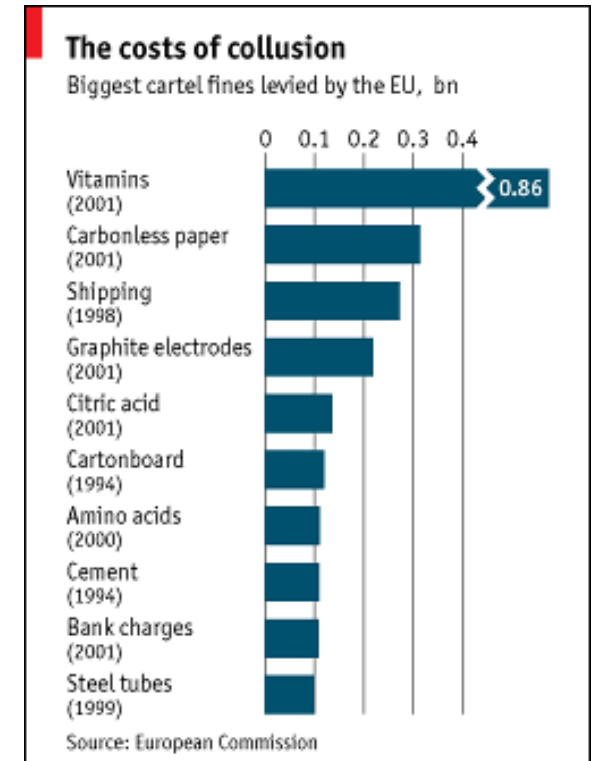
- Often directly government controlled

What causes monopolies?

5. Cartel

- Collusion in order to restrict output and sell for higher price
- Illegal
- Unstable due to incentives to break the rules

Examples



What causes monopolies?

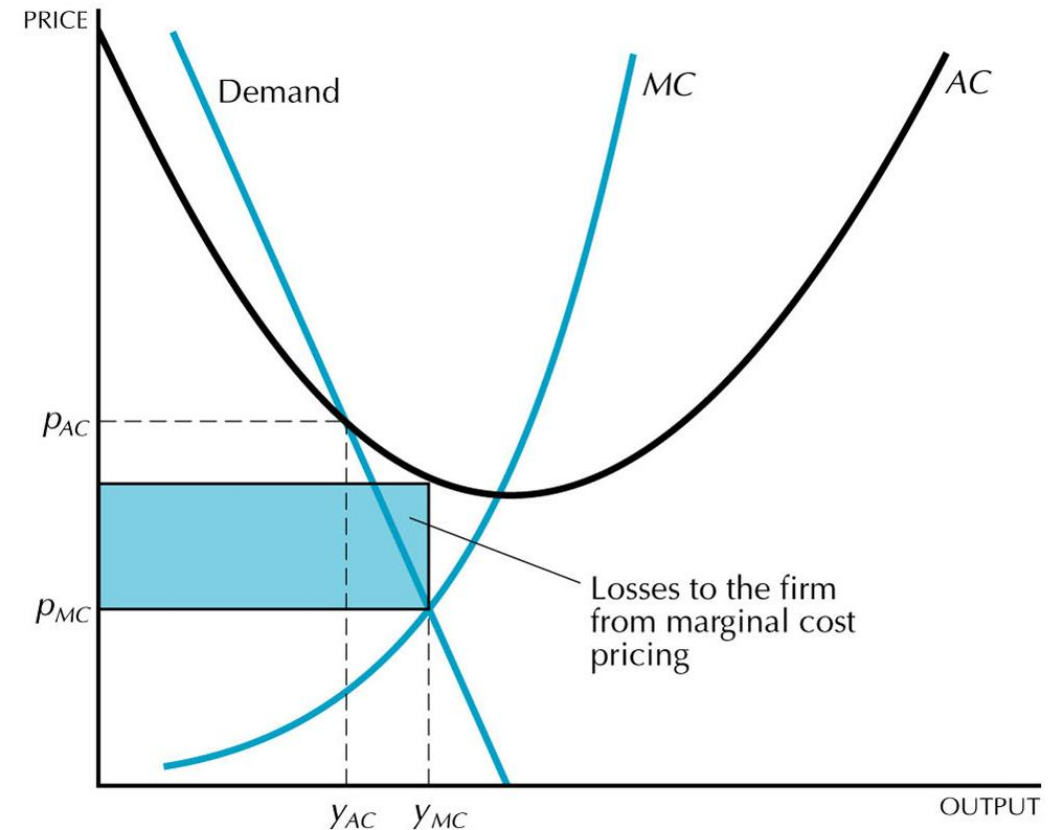
5. High costs of entry

- Large costs of entry (e.g. loyalty of customers)
- The incumbent may convince potential entrants it would cut prices if they try to enter → uncertainty exacerbates the effects of the entry costs
- The costs can be high (for example) due to
 - Costs of fixed capital (e.g. semiconductor fabs)
 - Costs of training personnel (e.g. nuclear powerplants)
 - Costs of R&D (e.g. pharmaceuticals)
 - Regulatory costs (e.g. pharmaceuticals)



Regulation of natural monopoly

1. Set price equal to marginal cost
 - Monopoly may make negative profit
 - Example: public utilities (electricity, gas, water)
 - Problems:
 - if negative profits → need for subsidies
 - regulator may not know the marginal and average costs
 - → regulatory capture
2. Intersection of the average costs and market demand
 - Monopoly does not make loss
 - The service provided to all who want to pay for it
 - Problems:
 - inefficient (output smaller than optimal)
 - again regulator may not know the actual costs



Group discussion

Please get in groups of 3.

1. A pharma firm has a patent over a drug. The demand for the drug is more elastic in Europe than in the US. Where will the firm set higher price? And what could be a reason demand in Europe is more elastic?
2. You have been asked to advise a government to investigate whether a railway signalling manufacturer is a (quasi)monopoly. What indicators will you look at and why?
3. Railways are a classical example of a natural monopoly. How are railways regulated in your country? Are infrastructure providers and train operators separated, and if so, why? Are there differences between regulation of inter-city and regional train lines?

